Geoderma 445 (2024) 116882

Contents lists available at ScienceDirect

Geoderma

journal homepage: www.elsevier.com/locate/geoderma

Measurement and prediction of gas permeability function for biochar-amended rooted soils

X. Huang [a], W. Cai [b], S. Bordoloi [c,*]

[a] Department of Geotechnical Engineering, School of Civil Engineering, Southwest Jiaotong University, Chengdu 610031, China
[b] Department of Civil and Environmental Engineering, Rowan University, New Jersey 08028, United States
[c] Department of Civil Engineering, Aalto University, Espoo, Finland

ARTICLE INFO

Handling Editor: L. Morgan Cristine

Keywords:
CO_2 permeability
Root-soil interaction
Ecological restoration
Biochar

ABSTRACT

Biochar amendment have been explored recently for improving hydro-mechanical properties of landfill cover system and enhancing the vegetation for the ecological restoration of post closure landfills. Existing studies have predominantly focused on the hydro-mechanical properties of biochar-amended rooted soils and yet few studies have comprehensively investigated their gas permeability functions in response to unsaturated conditions. The objective of this study is to experimentally and theoretically investigate the coupled effects of biochar and root intensity on CO_2 gas permeability in unsaturated soil. Biochar-amended rooted soil was prepared by growing *Chrysopogon zizanioides* on granite residual soils with biochar at a mas ratio of 5 %. Gas permeability and unsaturated soil properties (i.e., suction and degree of saturation) were monitored simultaneously. Due to the presence of roots, the maximum gas permeability, hydraulic conductivity, and suction increased by 200 %, 600 %, and 50 %, respectively, compared with bare soil. This was attributed to preferential flow, micropore increasement and negative pressure induced by root growth. However, biochar addition decreased the maximum gas permeability and hydraulic conductivity by 21 % and 33 %, respectively. This was attributed to the pore filling function of biochar, increased capillarity due to intrapore of biochar, and presence of surface functional groups in biochar promoting CO_2 adsorption. A newly developed model was proposed to predict gas permeability with respected to unsaturated soil properties. It was revealed that the gas permeability function of rooted soils showed a strong dependence on the measured root length density. This study highlights the significance of adopting biochar in mitigation of gas emissions in vegetated landfill cover system.

1. Introduction

Geo-environmental infrastructures (e.g., landfill cover system, hydraulic barriers for mine tailings), have been widely promoted as suitable strategy for environmental management of degraded soils. Emphases is focused towards reducing or mitigating greenhouse gas emissions from landfills (carbon dioxide, CO_2 gas) (Gill et al., 2007). Hydraulic barriers with vegetation are a common strategy to protect the waste layer from infiltration as well as for ecological restoration of these landfill sites (Tavanti et al., 2023). Vegetation induces hydrological changes in vadose zone due to combined effects of transpiration, soil pore size distribution due to root occupancy, root decay and root exudate secretion (Bordoloi and Ng, 2020). The gas permeability of unsaturated vegetated soil is extensively investigated (Garg et al., 2019). Overall, it is observed that at high suctions there is increase in gas permeability of desiccated soil which exacerbates release of the greenhouse gas (e.g., CO_2 and methane) from the underground, in particular from landfills. From the perspective of unsaturated soil mechanics, gas permeability is primarily governed by the soil pore network. Essentially, gas permeability is attributed to the macroporosity (Cai et al., 2022), characteristic of desiccation cracking (Zhao and Santamarina, 2020). Therefore, it is essential to regulate the soil porosity and the associated gas permeability, to ensure the serviceability of geo-environmental infrastructure susceptible to greenhouse gas emissions.

Concerns have been raised on the serious gas emission issue in landfill cover systems. Typically, landfill is the most common practice to dispose municipal solid wastes (Shaikh et al., 2019), which are biochemically decomposed over time generating greenhouse gases (e.g., CO_2, CH_4, N_2O etc.). Even though gas exhaust systems are employed in the cover system, it has been reported that high CO_2 flux into

* Corresponding author.
E-mail addresses: XilongHuang2022@my.swjtu.edu.cn (X. Huang), caiwei58@students.rowan.edu (W. Cai), sanandam.bordoloi@aalto.fi (S. Bordoloi).

https://doi.org/10.1016/j.geoderma.2024.116882
Received 16 January 2024; Received in revised form 1 April 2024; Accepted 7 April 2024
Available online 17 April 2024

atmosphere occurs from the top vegetated layer of soil at high suctions (Ni et al., 2019). It should be noted that, after a landfill site has been closed, the landfill cover will be vegetated for aesthetical and ecological purposes. However, roots might induce changes in soil pore structure, exacerbating the emission of greenhouse gases in landfill. Ni et al. (2019) found that a grass species (*Cynodon dactylon*) increases the gas permeability up to one order of magnitude than that in bare soil. According to Kaza et al. (2018), the emission of CO_2 will reach 2.6 billion tons from landfill by 2050 if no improvements are made, which may lead to significant global environmental and climate issues. From engineering perspectives, a reduction of gas permeability in the landfill cover is desired to minimize the greenhouse gases emission into the atmosphere. In the past few decades, biochar, as an environmentally friendly sustainable soil amendment, have been utilized to improve hydrological functions including reduce the gas permeability (Sarmah, 2021). The biochar addition induced changes in soil pore structure can affect gas migrations in soil. Garg et al. (2019) found that the gas permeability of biochar amended soil was smaller than that of sand clay mixture by around 50 %. Similar results were also observed for biochar-amended clay soil without vegetation (Wong et al., 2016). Garg et al. (2020) also revealed that the addition of 5 % biochar in soil was optimum in decreasing gas permeability at a relatively low suction range (<200 kPa). Moreover, biochar further adsorbs CO_2 in the unsaturated state through either physisorption in the micropores, forming Van der Waals bonds with the available aromatic rings and Lewis's acid-Lewis's base interaction between available N-functional and O_2 functional bonds (refer Fig. 1a; Shafawi et al., 2021). Therefore, biochar as a soil amendment has the potential to minimize major greenhouse gas such as CO_2 in landfill cover engineering.

While the soil permeability has been investigated under the impact of biochar addition or vegetation only (Garg et al., 2019; Ni et al., 2019), the gas permeability of biochar-amended vegetated soils is barely investigated and understood yet. The coupled effects of vegetation and biochar on gas permeability are ambitious since the vegetation and biochar have contradictive effects on the gas permeability. Furthermore, biochar amendment results in a benefit of root proliferation and a change of root architecture (Abiven et al., 2015). This indicates the complexity of the soil-root-biochar interactions in the vadose zone. To tackle this issue, the authors hypothesized that the gas permeability of the biochar-amended rooted soil was associated with the root length

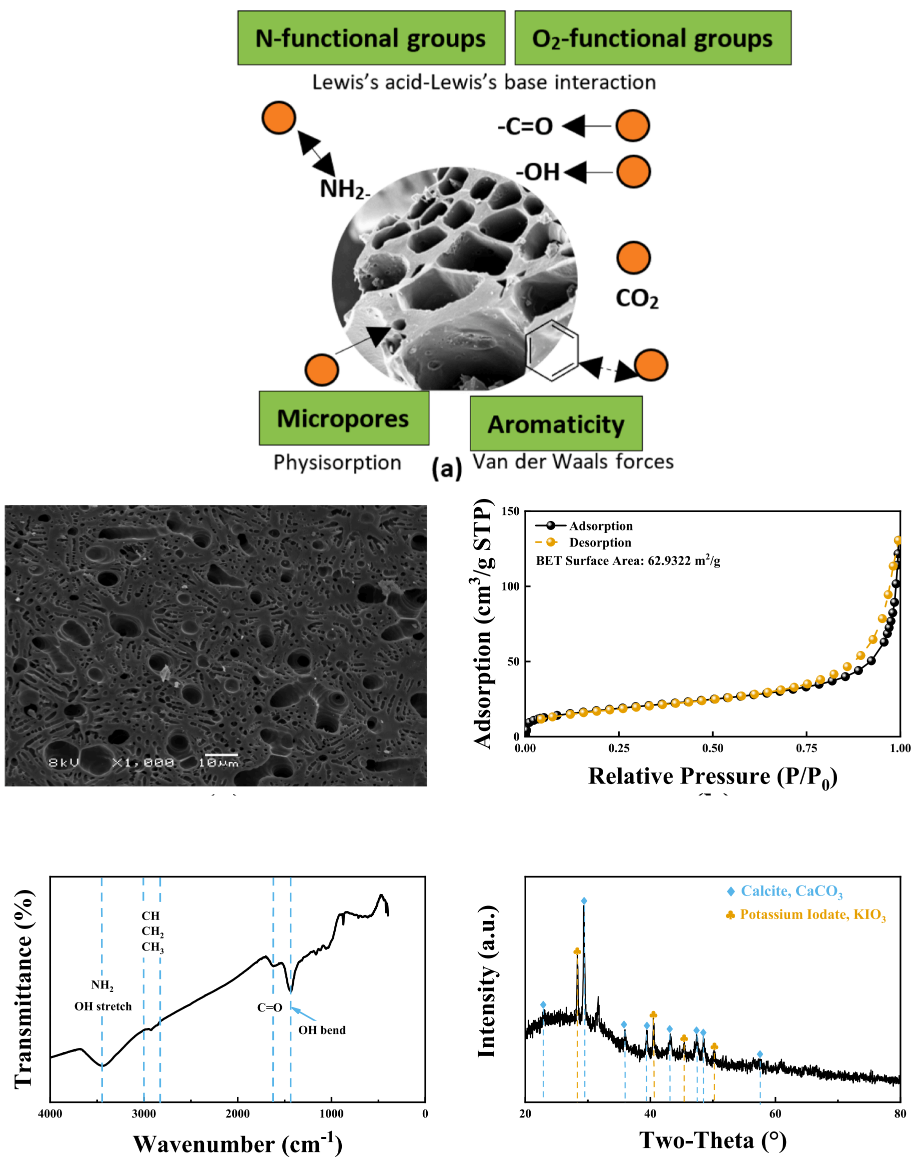


Fig. 1. **(a)** Schematic diagram exhibiting mechanism of CO_2 adsorption by biochar **(b)** Surface morphology of biochar; **(c)** Brunauer-Emmett-Teller (BET) analysis of biochar; **(d)** Fourier Transform Infrared (FTIR) spectra of biochar; **(e)** X-ray Powder Diffraction (XRD) of biochar.

density in the current study. An experimental and theoretical study was conducted to investigate the CO_2 permeability of biochar-amended rooted soils at various degrees of saturation.

Therefore, the main aim of this study was to investigate the coupled effects of biochar addition and root intensity on CO_2 gas permeability. A series of biochar-amended rooted soils were prepared. Water retention tests and gas permeability measurement were performed along with the temporal monitoring of the plant growth. Unsaturated responses (i.e., matric suction and volumetric water content) were monitored continuously. CO_2 permeability of soils was measured during the desiccation process, encompassing expected ranges of suction under field conditions. The saturated hydraulic conductivity of soils along with the root architecture was evaluated at the end of the test.

2. Materials and methods

2.1. Basic properties of soil and biochar

The biochar used in the current study was made from the carbonization of water hyacinths at the temperature of 600 ℃ under zero-oxygen conditions (Keiluweit et al., 2010). Numerous intrapore on the biochar surface were formed during the pyrolysis process, which was captured by the scanning electron microscope (SEM), and the micrograph is proffered in Fig. 1**b**. This was because the thermal decomposition of organic matter (e.g., cellulose and hemicelluloses) could reorganize the solid matrix to make up porous structure (Lee et al., 2013). The specific surface area of biochar was measured by Brunauer-Emmett-Teller (BET) analysis. The nitrogen adsorption/desorption isotherm of biochar was fitted by type II isotherm (Fig. 1**c**), indicating the prolific micropores and macropores on biochar. The presence of micropores can lead to sites for physisorption of CO_2 gas molecules under unsaturated conditions. The high specific surface area of biochar (62.93 m^2/g) indicated its heroic capacity to absorb water. Fig. 1**d** presents the surface functional groups of biochar characterized by Fourier Transform Infrared (FTIR) analysis. Hydrophilic functional groups, such as amido (–NH_2) and hydroxyl (–OH) were observed distinctly which would help in water retention. Presence of –NH_2, –OH and carboxyl C = O groups have strong affinity for CO_2 gas and can adsorb them through Lewis's acid-base interaction. X–ray diffraction (XRD) analysis was conducted to identify possible crystalline structures on biochar, as shown in Fig. 1**e**. In biochar particles, the presence of calcite, potassium iodate, and cellulose was confirmed by X-ray diffraction analysis. The broad peak in XRD spectra at 2θ values between 20 and 35 may demonstrates that cellulose evolved to turbostratic crystallites at charring temperatures above 400 ℃ (Keiluweit et al., 2010). The physiochemical characteristics of biochar are summered in Table 1.

The bare soil (BS) texture consisted of around 23 % coarse sand, 37 % medium sand, 21 % fine sand, 17 % silt, and 2 % clay, which was classified as clayey sand (SC) by following the standard of ASTM D2487-17 (2020). The soil was composited evenly with biochar, where the mass ratio of dried biochar was 5 % (BAS5%) having been showed remarkable potential in the reduction of gas emission and the promotion of plant growth (Garg et al., 2019; Guo et al., 2023). The optimum water content and the maximum dry density of BS were found to be around 12 % and 1.94 g/cm^3, respectively. Biochar would increase the optimum water content to around 13 % while decreasing the maximum dry density by 1 %. The geotechnical properties of the bare soil and biochar-amended soil are summarized in Table 2.

Table 1
Production conditions and chemical properties of biochar.

Index	Value
Feedstock source	*Water hyacinth*
Pyrolysis temperature (℃)	600
Particle size distribution (%)	
0.075–0.425 mm	30.48
$< 75 \times 10^{-3}$	68.52
$< 2 \times 10^{-3}$ *mm*	1.00
pH	9.25
Specific gravity	0.80
Surface area (m^2/g)	62.93
Median pore width (nm)	13.81

2.2. Experimental setup and testing procedure

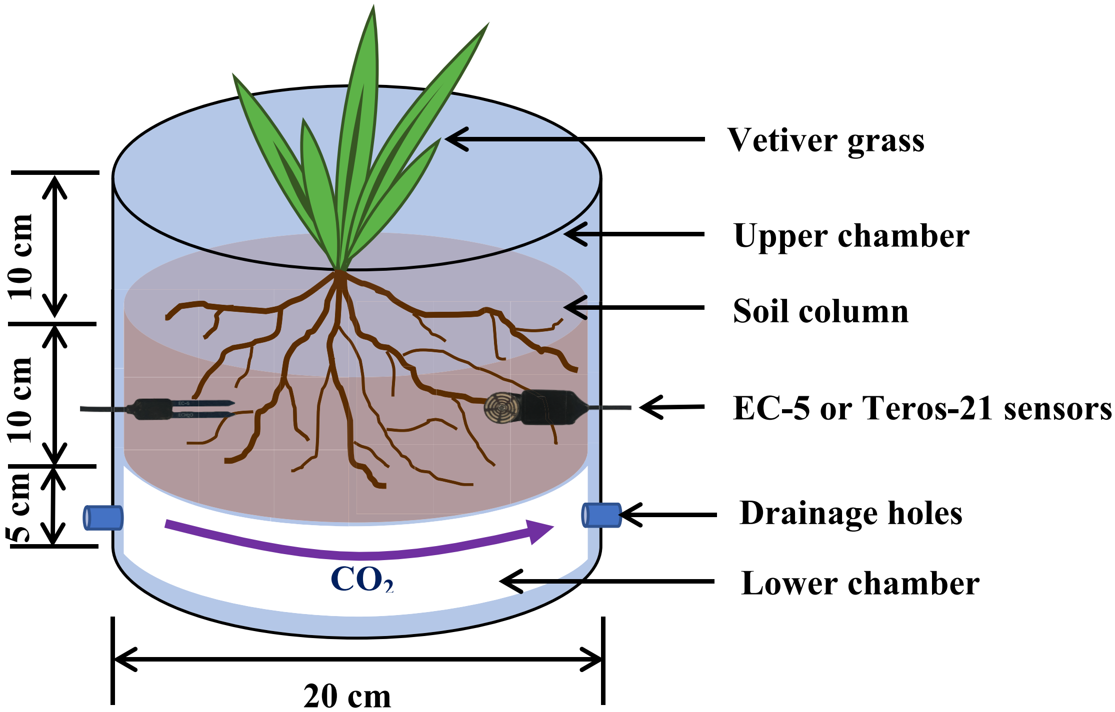
Fig. 2 indicates a simple device for performing the water retention test and gas permeability measurement and growing vetiver grass (*Chrysopogon zizanioides*) on the soil. This species is commonly used for ecological rehabilitation and slope stabilization for infrastructures in many Asian countries. Before transplantation, vetiver grass individuals were trimmed to a uniform size with a shoot length of 25 ± 5 cm and root depth of 4 ± 1 cm. A Poly Vinyl Chloride (PVC) cylindrical pot was divided into two chambers. The upper chamber, with a diameter of 20 cm and a height of 20 cm, was used to place the soil sample. A soil column was manufactured with a height of 10 cm in the upper chamber. The soil was compacted statically in three layers along the longitudinal axis of the mold, controlled by 80 % of compaction degree. A volumetric water content sensor (EC-5) and a matric suction sensor (Teros-21) were installed at the middle depth (5 cm) of the soil column. The lower chamber supplied a constant pressure to allow gas to flow through soil column. Drainage holes were made at the bottom of the lower chamber for free drainage and gas input.

Three vetiver grass individuals transplanted in the soil column represented soil layers with low vegetation density. Correspondingly, nine vetiver grass individuals transplanted in the soil column simulated soil layers with high vegetation density. Two types of vegetated soil, namely bare soil with low vegetation density (BSVL) and bare soil with high vegetation density (BSVH), were prepared. The nutrient effects of biochar on plant growth in high vegetation density was negligible due to the plant competition issue (Ni et al., 2019). Hence, biochar amended soil with low vegetation density (BASVL) was created. In total, six soil columns were constructed, including two for the BSVL, two for the BSVH, and two for the BASVL (Table3). One bare soil column (BS) was regarded as a control group in the current test.

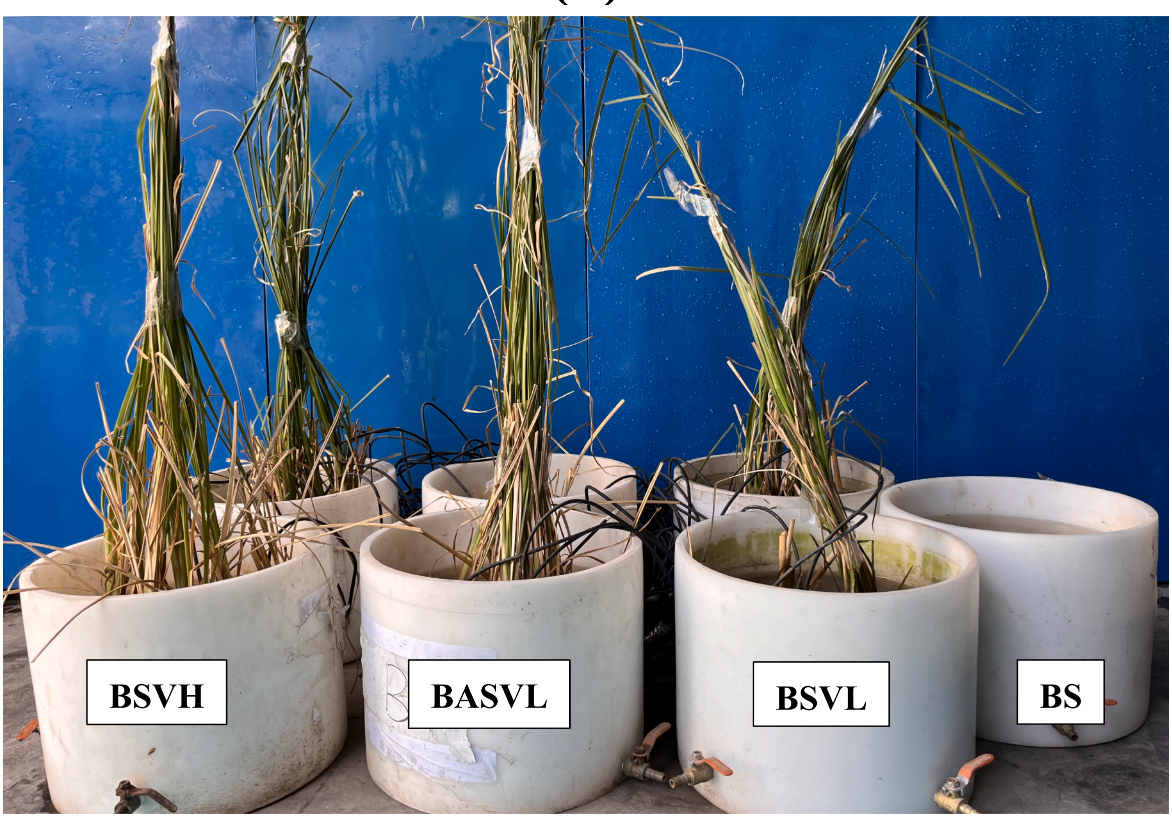
After transplantation, to ensure the interaction between roots and soil sufficiently, vetiver grasses were left to grow for 40 days with sufficient illumination and irrigation. Then, the seven columns were ponded with a controlled constant head of 25 cm for 8 h, and the saturated hydraulic conductivities of each column were measured when the outflow at the drainage holes were at a constant rate. Subsequently, all columns were left in the outdoor for evapotranspiration until the volume water content was stable. During the evapotranspiration process, gas permeabilities were measured at an interval of 8 h. For measuring gas permeability of samples, CO_2 gas was supplied by a compressed gas

Table 2
Index properties of bare soil and biochar-amended soil.

Soil properties	BS	BAS5%
Particle size distribution (%)		
Coarse sand (2–4.75 mm)	23.11	21.95
Medium sand (0.425–2 mm)	37.28	35.42
Fine sand (0.075–0.425 mm)	20.84	21.32
Silt ($< 75 \times 10^{-3}$ *mm*)	17.00	19.58
Clay ($< 2 \times 10^{-3}$ *mm*)	1.77	1.73
Maximum dry density (g/cm^3)	1.94	1.92
Optimum moisture content (%)	12.02	12.56
Specific gravity	2.59	2.59
Calculated void ratio	0.69	0.71
Atterberg limit (%)		
Liquid limit (LL)	42.4	
Plastic limit (PL)	26.0	
Plastic index (PI)	16.4	



(a)



(b)

Fig. 2. **(a)** Schematic diagram of the experimental device and **(b)** photo of soil columns.

Table 3
Experimental plan.

Sample ID	Vegetation density	Biochar	Degree of Compaction
BS	None	None	80 %
BSVL	Low	None	80 %
BSVH	High	None	80 %
BASVL	Low	5 %	80 %

Note: Three clumps of vetiver grasses planted in the soil column are labeled as low vegetation density, and nine clumps of vetiver grasses planted in the soil column are labeled as high vegetation density.

cylinder to flow though the lower chamber and soil column steady. Gas dissolution can be reasonably ignored in this experiment considering that the applied pressure was only slightly higher than the atmospheric pressure. The gas pressure at the lower chamber presented the pressure difference across the soil column. Flow rate (Q, m^3/s) and gas pressure (ΔP, Pa) can be measured and thus gas permeability was determined by Darcy's Law (Detailed introduction in Section 2.3.1). In the time of the gas permeability measurement, one drainage hole received gas flow to produce gas pressure and the other connected a digital pressure sensor to determine pressure in the pressure chamber. CO_2 was allowed to flow through the soil column under a constant pressure gradient supplied by the lower chamber. Meanwhile, the degree of saturation (Sr) and suction (ψ) were recorded simultaneously. The shoot lengths of each grass were monitored every day during the experiment.

After the test, entire root systems were excavated and washed free of soil with tap water. Dead roots in the system, recognized by sight, touch and flotation, were discarded. Screening results showed ideal root growth in all experimental groups with living stele, bright color, and elasticity, almost without decay roots. All living roots separated from each soil sample mass were scanned around 360° with 600 dpi resolution. Root lengths and area at any depth within root zone were estimated in each sample using an image analysis conducted by Image J. Root area index (*RAI*) and Root length density (*RLD*) were calculated for specific interval of soil layer.

2.3. Theoretical consideration

2.3.1. Gas permeability

Gas permeability represents the property of soils that reflects the capability of gas to move through the soil body in the presence of a pressure gradient. Gas permeability (k_a, m^2) was calculated by an equation derived from Darcy's Law as follows (Juca and Maciel, 2006):

$$k_a = \frac{\mu Q L}{\Delta P A} \tag{1}$$

where μ is the absolute viscosity of gas (e.g., for CO_2 at room temperature and atmosphere, $\mu = 15.0 \times 10^{-6} \mathrm{N{\cdot}s/m^2}$); Q is the average volumetric flow rate of fluid (m^3/s); L is the length of the flow path (m); ΔP is pressure difference across the porous medium (Pa); and A is the cross-sectional area perpendicular to the flow direction (m^2).

2.3.2. Soil-water characteristic curves (SWCC)

The SWCC describes the association between the water content and the soil suction, which is widely interpreted by van Genuchten (1980) model.

$$Sr = \left(\frac{1}{1 + (\alpha\psi)^n}\right)^m \tag{2}$$

where Sr is the degree of saturation; ψ is the matric suction (kPa); α is the inverse of the air entry value (kPa^{-1}); m and n are the fitting parameters (i.e., $m = 1 - 1/n$).

In practice, the correspondences between Sr and ψ are not unique for a specific soil due to the variation of the void ratio. Gallipoli et al. (2003) modified Eq. (2) to incorporate volume changes, by making the parameter α dependent on specific volume (v):

$$\alpha = a(e)^b \tag{3}$$

where e is the soil void ratio; a and b are soil constants.

2.3.3. Gas permeability function

The maximum gas permeability in vegetated soils can be expressed in a normalized form (k_{ra}):

$$k_{ra} = \frac{k_d}{k_{d,BS}} \tag{4}$$

where k_d is the maximum gas permeability of certain soil (m^2); and $k_{d,BS}$ is the maximum gas permeability of bare soil (m^2).

Brooks and Corey (1964) presented the following expressions for k_a correlated with S:

$$\begin{cases} k_a = 0.0 & for\ \psi \leqslant \psi_b \\ k_a = k_d(1-S)^2\left(1 - S^{(2+\lambda)/\lambda}\right) & for\ \psi \geqslant \psi_b \end{cases} \tag{5}$$

Eq. (6) also can be expressed in term of ψ:

$$\begin{cases} k_a = 0.0 \\ k_a = k_d\left[1 - \left(\frac{\psi_b}{\psi}\right)^{\lambda}\right]^2\left[1 - \left(\frac{\psi_b}{\psi}\right)^{2+\lambda}\right] \end{cases} for\ \psi \leqslant \psi_b\ for\ \psi \geqslant \psi_b \tag{6}$$

where ψ_b is the air-entry value of soils (kPa); λ is a fitting parameter.

Another predictive model of the k_a of rooted soil was proposed by Feng et al. (2023), which was expressed as:

$$\begin{cases} k_a = \alpha_r(1-S)^{X_r}\eta^{X_r} \\ \alpha_r = \alpha_a + 3.66 \times 10^{-8}R_V - 1.47 \times 10^{-10} \\ X_r = X_a + 7.23R_V - 0.03 \end{cases} \tag{7}$$

where α_r is a constant reflecting the pore tortuosity-connectivity; X_r is a power law exponent related to water blockage on gas movement; η is the soil porosity; R_V is the volume of root per unit soil volume; α_a and X_a are constants observed through the best-fitting of the data of the bare soil using Eq.(7).

2.3.4. Root traits

Root length density (*RLD*, cm/cm^3) is a significant indicator to estimate the intensity of soil exploration, defined as the total root length per unit volume soil:

$$RLD = \frac{RL}{VS} \tag{8}$$

where RL is the total root length in a soil sample (cm); VS is the volume of the soil sample (e.g., in this study, $VS = 6605cm^3$).

Root area index (*RAI*, m^2/m^2) estimates the lateral extent from the center of plant that roots can explore in soil, determining by the following formulation:

$$RAI = \frac{RA}{SA} \tag{9}$$

where RA is the total root area for a given root depth (m^2); SA is the plan cross-sectional area of soil (m^2).

2.3.5. Estimation of hydraulic conductivity

The relative saturated hydraulic conductivity (k_{rw}) can be written as follow:

$$k_{rw} = \frac{k_s}{k_{s,BS}} \tag{10}$$

where k_s is the saturated hydraulic conductivity of certain soil (cm/s); $k_{s,BS}$ is the saturated hydraulic conductivity of bare soil (cm/s).

The Konzeny-Carman (KC) equation relates the saturated hydraulic conductivity k_s to void ratio e, which can be expressed as (Mitchell & Soga, 2005):

$$k_s = \left(\frac{1}{k_0 T_u^2 S_0^2}\right) \bullet \left(\frac{\gamma_w}{\mu}\right) \bullet \left(\frac{e^3}{1+e}\right) = C \bullet \left(\frac{e^3}{1+e}\right) \tag{11}$$

where k_0 is a pore shape factor; μ is the dynamic viscosity (Pa·s); T_u is a tortuosity of the flow path; S_0 is the wetted surface per unit volume of particles; C is a constant calibrated based on a bare soil if there are not decay roots in the vegetated soil (Ni et al., 2019).

The van Genuchten (VG) hydraulic model predicts the unsaturated hydraulic conductivity (k_w, cm/s) from k_s and ψ, which combines by van Genuchten model (1980) as:

$$k_w = k_s \bullet \frac{\left\{1 - (\alpha\psi)^{n-1}\left[1 + (\alpha\psi)^n\right]^{-m}\right\}^2}{\left[1 + (\alpha\psi)^n\right]^{0.5}} \tag{12}$$

2.4. Statistical analysis

Statistical analysis was employed to examine the significance level of influence for biochar and vegetation addition on unsaturated soil properties (i.e., e, α, and k_d). The experimental results were analyzed by a two-way analysis of variance (ANOVA) to assess the statistical significance level, as summarized in Table 4. The results of F–distribution validated the reliability of the interpretation. Statistically significant results were certified by p-values less than a predetermined significance level (i.e., 0.5).

3. Results and discussion

3.1. Evolution of shoot height and root of vetiver grass

Temporal variations of vetiver grass shoot height in different soil media (with and without biochar) are presented in Fig. 3. For each group, every individual of vetiver grass shoot height was measured, summed, and divided by the amount of the individuals to capture the average value, which indicated by points. The difference of shoot heights between BSVL and BSVH were negligible. Shoot heights of BASVL remained higher around 20 % than that of the vetiver grass without any treatment. The $–NH_2$ and KIO_3 in the biochar, as shown in Fig. 1, can serve as fertilizer which can be absorbed by vetiver grass. The available N and K (e.g., NH_4^+ and K^+) can increase biomass accumulation, promote photosynthesis, and advance the movement of water, nutrients, and carbohydrates in plant tissue (Wang et al., 1996). The high pH of the biochar, shown as 9.25 in Table 1, would prevent the nutrient to leach (Lehmann et al., 2011). In addition, the porous structure (Fig. 1**b**), high surface area (Fig. 1**c**) and hydrophilic functional group (NH_2, –OH) (Fig. 1**d**) of biochar could improve the water and nutrients storage capacity, facilitating easier root water and nutrients uptake for plant growth (Joseph et al., 2021). The improvement of biochar in plant growth was also reflected in the root architecture development. Fig. 4 demonstrates the *RAI* distribution with soil depth for the BSVL, BSVH and BASVL. The *RAI* didn't decrease until reaching a maximum value at around 0.2 of normalized depth, whose morphology were in parabolic shape along the depth as similar as found by shao et al. (2017). It was observed that the proliferated root in BASVL laterally increased by 44 % and 23 % in comparison to BSVL with respect to the normalized depth at 0.2 and 0.4, respectively. In contrast, the *RAI* of BASVL reduced around 45 % and 82 % with a comparison of BSVL at the normalized depth at 0.8 and 1.0. Although the biochar facilitated higher root growth, the root architecture with and without biochar addition were significantly different. High availability of immobile nutrients due to biochar addition facilitated the proliferation of primary and secondary lateral roots (Peng et al., 2010). This would have implication in development of preferential flow of gas as tortuosity will be higher at shallow depths. The negative effect on root vertical proliferation in the biochar treated soil may be also due to bad aeration in lower parts of the subsoil as consequence of high-water retention (Bruun et al., 2014). It was found that the specific gravity altered to 2.59 from 2.69 after biochar application, as indicated in Table 2. The reduction of soil density would create wider and additional pores to promote root proliferation (Bruun et al., 2014). Overall, biochar addition increased soil fertility, absorption of water and nutrients, and soil porosity, which can facilitate the proliferation of root in compacted soil, typical to landfill covers. In addition, lower vertical root penetration in biochar treated soil will thereby protect the integrity of the layers in landfill cover systems.

Table 4
Two-way ANOVA analysis on the influence of biochar and vegetation amendment.

Dependent variable	Biochar amendment		Vegetation amendment	
	F-ratio	p	F-ratio	p
e	601.24	8.17×10^{-9}	4514.66	2.68×10^{-12}
α	949.98	1.33×10^{-9}	1104.21	7.34×10^{-10}
k_d	128.67	3.29×10^{-6}	1164.26	5.95×10^{-10}

3.2. Impact of biochar and root on SWCC

Soil water characteristic curves (SWCC) of bare soil, rooted soil, and biochar-amended rooted soil are shown in Fig. 5. The *Sr* of the soil started to decrease until its ψ reached to the air entry value. Subsequently, the *Sr* maintained zero with the increasement of ψ. SWCCs were fitted using the model of van Genuchten (1980), and the calibrated parameters were summarized in Table 5. It was found that the S_r increased with vegetation density. At any specific suction, BSVH had higher degree of saturation than the BS and BSVL. This was attributed to the reduction of soil pore space caused by the root occupancy, resulting in decrement of pore diameter, consequently increased suction at a specific S_r (Lu et al., 2020). In addition, root water absorption could improve soil water retention capacity by providing additional negative pressure (Xiao et al., 2023). It also should be noted that the S_r of BASVL was higher than that of BSVL. Compared with BSVL, the addition of biochar increased the air entry value (AEV) by around 6 % (see value $\boldsymbol{\alpha}$ in Table 5). Indeed, the existence of biochar facilitated roots growth to cause a further increasement of micropores, resulting in enhancement of capillarity. The porous and hydrophilic characteristic of biochar also would enhance the capillarity in soil (Cai et al., 2022). A similar improvement in air entry value caused by biochar addition was also observed by Cai et al. (2022). Interestingly, opposite results were found by Jeffery et al. (2015) who showed the effect of biochar on SWCC was not significant. The different results were caused by the different wettability (i.e., hydrophilic, and hydrophobic) of biochar used in studies. Biochar used in this study was produced from water hyacinth, which was regarded as hydrophilic in previous study (Garg et al., 2020). On the contrary, the hydrophobic biochar would prevent water entering its interpore (Jeffery et al., 2015).

A change of void ratio (e) affected the inverse of air entry value (α) was illustrated in Fig. 6. Gallipolli's model (Gallipolli et al., 2003) demonstrated a considerable result in describing the relationship between α and e, with an R^2 was equal to 0.93. The α had an exponential raise with the increment of e, indicating the reduction of air entry value (AEV). Interestingly, although the BSVL and BASVL had a similar e, the α of BSVL was smaller around 6 % than that of BASVL. Biochar addition would facilitate root growth and hence caused the reduction of void ratio due to the root occupation. However, the intrapore of biochar would increase the void ratio in soil. The coupling influence of biochar and root increased the micropores while reducing macropores, resulting in enhanced capillary suction in the soil.

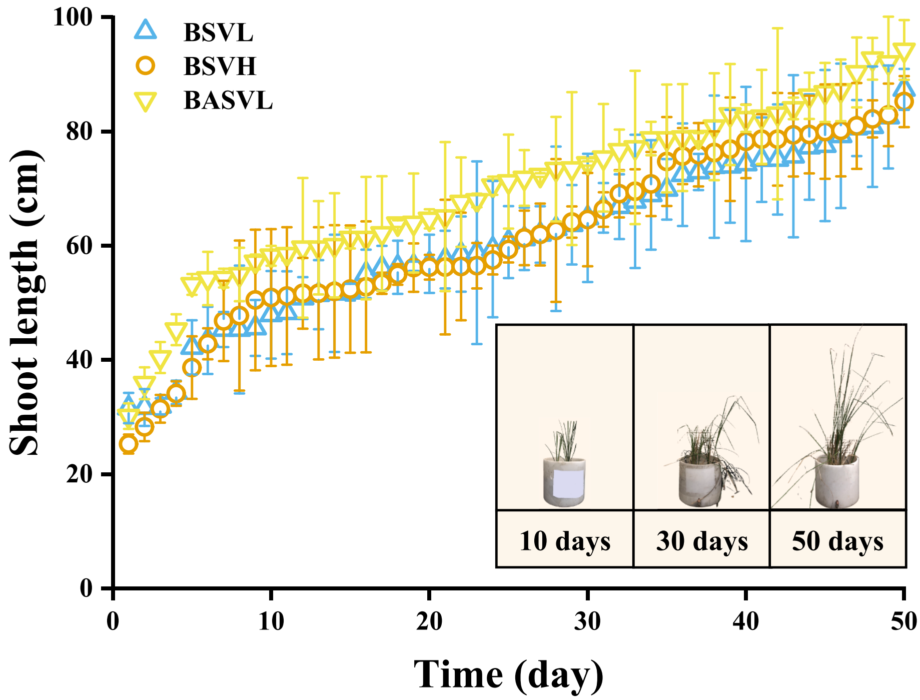


Fig. 3. Temporal variation of shoot height of vetiver grass on soils with and without biochar addition.

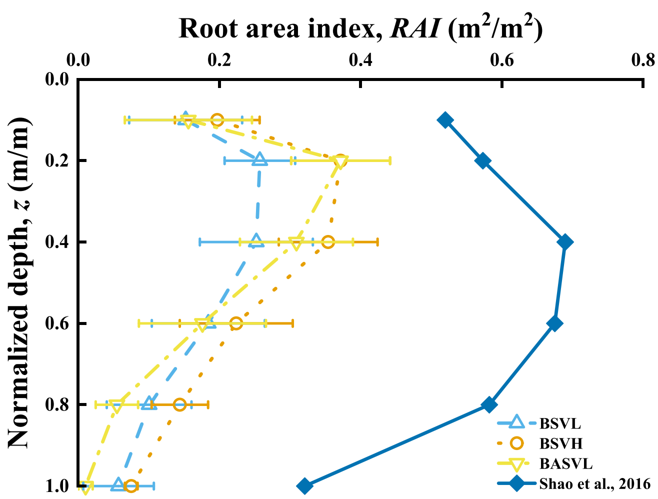


Fig. 4. Root morphology of vegetated soils with and without biochar addition.

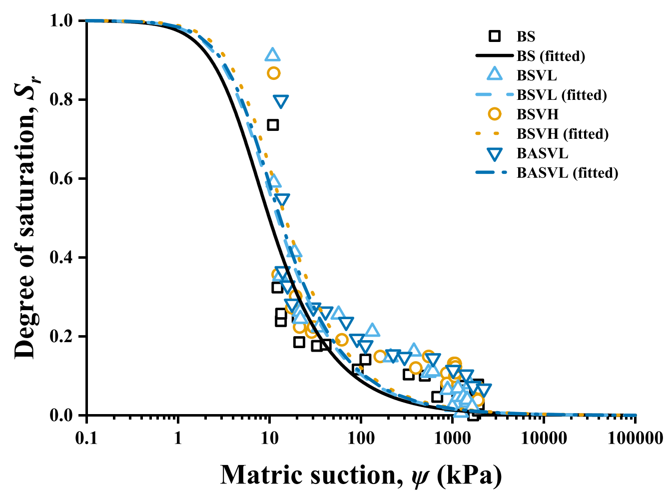


Fig. 5. Influence of biochar and root on SWCCs.

3.3. Effects of biochar and root on the permeability

Biochar-amended rooted soils showed the dependency of gas permeability (k_a) on the matric suction or degree of saturation, as indicated in Fig. 7. During the desaturation process, the k_a of the soil was maintained at its lowest value unit the air entry value was reached (i.e., gas breakthrough phenomenon) (Delahaye et al., 2002). Thereafter, the k_a of soils increased with a raise in matric suction and then stabilized at the maximum value (i.e., dry gas permeability, k_d). The gas permeability functions of biochar-amended rooted soils were fitted using the model of Brooks and Corey (1964). The fitting parameters were summarized in Table 6. The model showed a considerable fitting performance to capture the influence of the matric suction or degree of saturation on the gas permeability. The rooted soil had a higher gas permeability with increasing root intensity, for a given suction value beyond the air entry value. As compared to the bare soil, the k_d of the rooted soils with the low and high root intensities increased by 100 % and 200 %, respectively. This might be attributed to the macroporosity enlarged by the root proliferation (Hu et al., 2019). In contrast, Chen et al. (2023) found that the vegetation species and the root intensity had a positive impact on the gas permeability. The phenomenon was puzzling due to the existence of rhizosphere seems not to effect soil structure. Although roots can cause densification of adjacent rhizosphere soil, the existing pore in the rhizosphere would be enlarged (Bodner et al., 2014). As indicated in Fig. 8**a**, the rhizosphere is composed of roots, rhizosheaths

Table 5
Fitting parameters of SWCC model (van Genentech, 1980).

Specimen ID	α(kPa^{-1})	n	m	R^2
BS	0.21	1.80	0.44	0.73
BSVL	0.17	1.80	0.44	0.77
BSVH	0.14	1.80	0.44	0.72
BASVL	0.16	1.80	0.44	0.77

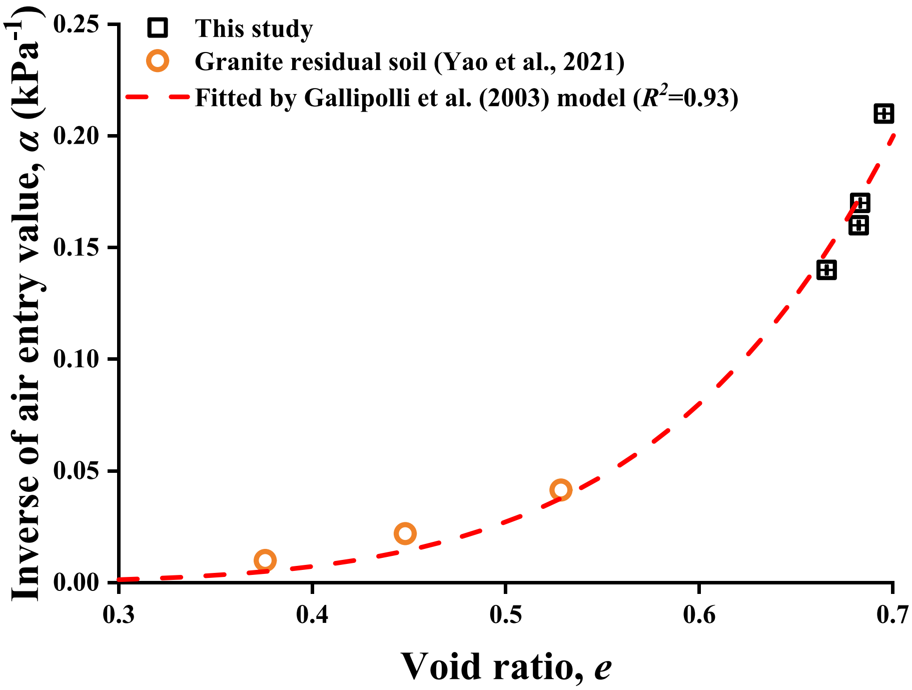


Fig. 6. Dependency of the air entry value of (biochar-amended) vegetated soils on void ratio.

and the soil around the roots, whose characteristics differ from those of rootles soil (Tarafdar and Jungk, 1987). During the penetration of root, the temporal increasement of root apex in diameter would reorganize the soil particles around the apex surface (Kolb et al., 2017). Soil particles would cling to roots through biotic glues, making porous coatings on the root (i.e., rhizosheath). The porous coatings along roots created a structured channel for preferential flow, whose flow tortuosity exhibited a reduced tendency compared to the pores found in the bulk soil (Koebernick et al., 2017). Air and water could flow preferentially through the restructure of pores in vegetated soil, which were critical in the transport of fluid in the unsaturated zone. Overall, the presence of roots would negatively affect the gas permeability function. However, due to the addition of biochar, the gas permeability of the rooted soil decreased. For the BASVL, biochar addition reduced the dry gas permeability by 21 % compared with BSVL. Similarly, decrease of k_d was observed in the addition of biochar at the 5 % and 10 % mass ratio for bare soil, which decreased by 33 % and 53 % respectively (Garg et al., 2019). This might be due to the pore clogged by biochar particles (Fig. 8**b**). The occupation of biochar in soil would reduce the pore size and soil pore connectivity, and hence, causing the reduction of k_a (Wong et al., 2016; Garg et al., 2019). The present of biochar would facilitate the root lateral growth, increasing the pore tortuosity. With the increasement of root volume affected by biochar, the soil would be further densified. The current findings suggest that the vegetation would increase the soil gas permeability, which might promote the emission of greenhouse gases. The greenhouse gas emission in the rooted landfill cover could be alleviated by biochar.

It should be noted that the gas permeability of soils is usually considered as dependent on soil porosity. The measured gas permeability with respect to the degree of saturation of biochar-amended rooted soils was predicted using the model proposed by Feng et al. (2023), as shown in Fig. 9. The model was calibrated based on the data of the bare soil and thereafter was applied to predict the gas permeability of biochar-amended rooted soils by changing the corresponding degree of saturation. The predicted results were presented in Table 7, including root mean square error (*RMSE*) and the corresponding predicted parameters. It was found that the model reasonably captured the gas permeability function of the bare soil, performing an ideal *RMSE* (12.67). However, the prediction underestimated the gas permeability of rooted soils, whose *RMSE* reached more than 75. This could be because the model cannot capture the influence of preferential flow induced by the root penetration on the gas permeability. Channels formed around live roots would capture subsurface gas and initiate a preferential flow, whereas the gas would bypass voids far aways form the roots and located in unsaturated areas (Ghestem et al., 2011). Therefore, the gas permeability of vegetated soil was dependence on root intensity and length strongly. Concerns should be paid on the influence of the root intensity on the hydrological properties of the soil.

Root intensity and length can be represented by root length density (*RLD*). Fig. 10 showed a simple model to predict the permeability of vegetated soil, whose values of k_{ra} and k_{rw} were related to the *RLD*. As expected, there was an increase in k_{ra} and k_{rw} with *RLD*. The increase can be described by a linear law (with $R^2 = 0.86$) and an exponential law (with $R^2 = 0.75$), respectively. Compared with slightly vegetated soil with 2.31 of *RLD*, highly vegetated soil with 5.56 of *RLD* can improve around 26 % of k_{ra} and 120 % of k_{rw}. The present of root in soil would induce the preferential flow and its effect would be stronger with increase of root density (Shao et al., 2017). Interestingly, biochar addition would promote the root growth in soil and inhibits the preferential flow phenomenon. As indicated in Fig. 10**a**, biochar increased *RLD* from 2.32 to 4.92 for slightly vegetated soil, whereas it decreased k_{ra} by around 10 %. Biochar application altered a root system to wider by alleviating deficiencies of soil water, which have been indicated in Fig. 4 (Bruun et al., 2014; Abiven et al., 2015). Preferential flow path may change from vertical to lateral directions due to the lateral penetration of root, which caused the decrease in permeability (Wang et al., 2018). The existence of biochar would block some parts of gas transmission channel, which would further decrease the gas permeability in soil (Garg et al., 2019). Fig. 10**b** presented the k_{rw} with regard to *RLD* predicted by Eq (10). As instructed by Ni et al. (2019), the parameter *C* was calibrated based on known *e* and k_s of a bare soil with a situation that there was no decay root in vegetated soil. It was predicted that a decrease trend in k_{rw} with an increase in *RLD*, which was contrary to the actual measurement results. In the view of Ni et al. (2019), preferential flow only occurred in the channels formed by dead or decaying roots, whereas living roots occupied pores to prevent fluid flow. However, the temporary increase

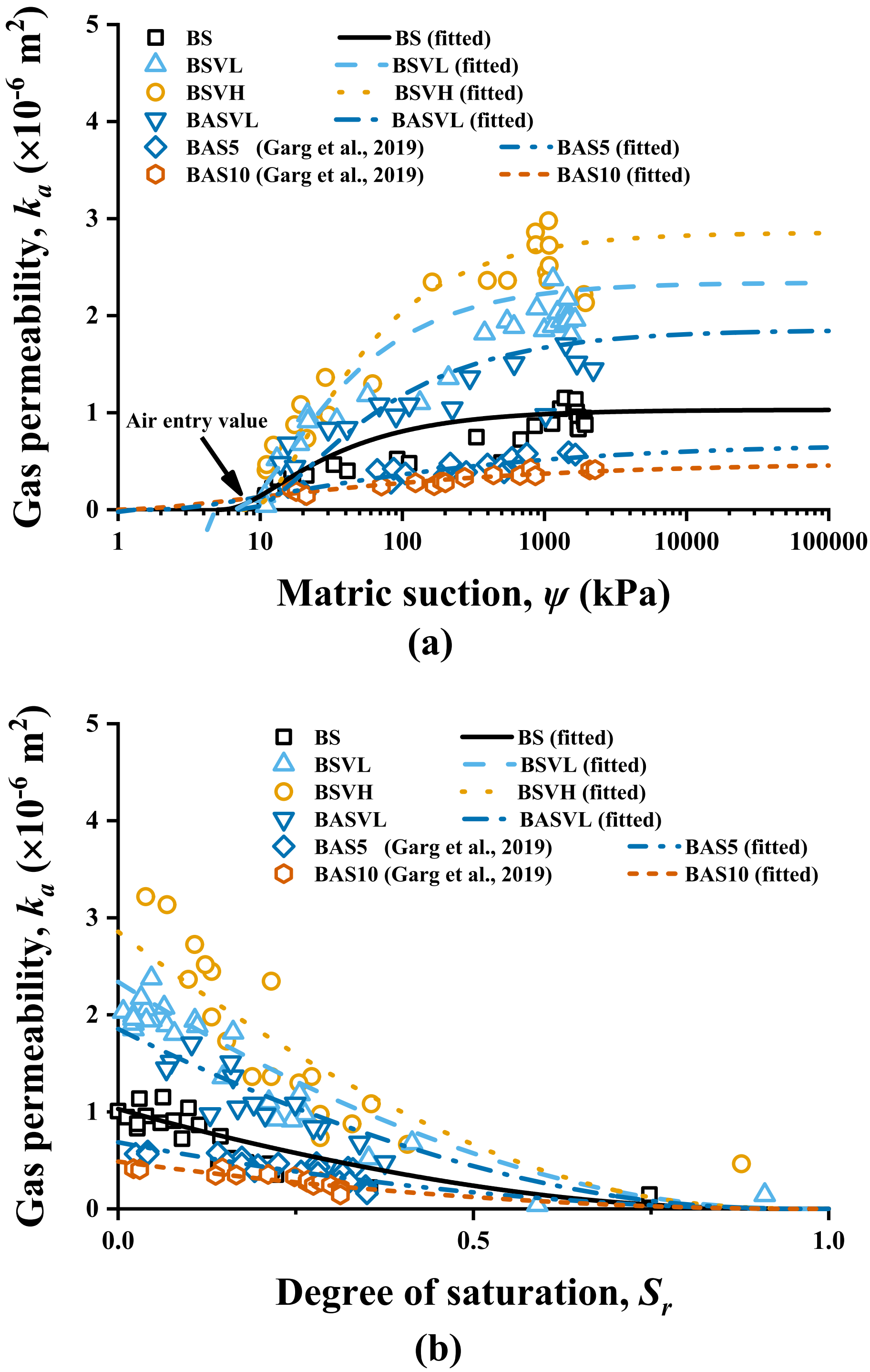


Fig. 7. Schematic representation of the modification of soil porous structure for **(a)** vegetated soils and **(b)** biochar-amended rooted soils.

Table 6
Fitting parameters of gas permeability functions (Book and Coney, 1964).

Specimen ID	k_d	λ	ψ_b	R^2
BS	102.91	0.72	4.90	0.74
BSVL	234.08	0.72	5.80	0.82
BSVH	285.67	0.72	7.50	0.90
BASVL	185.49	0.60	7.00	0.74
BAS5	68.56	0.31	1.52	0.75
BAS10	48.71	0.29	0.81	0.84

of root apex diameter would facilitate penetration through soil and soil particles redistribution (Ghestem et al., 2011). In addition, rhizosheaths adhered to roots in dry conditions would cause porous mantles around roots, which will create a structured channel for preferential flow (Ghestem et al., 2011). The underestimation of KC equation indicated that the effect of roots on soil permeability is complex and cannot be predicted solely by changes in porosity. In engineering, measuring root length density is often more convenient and economical than measuring porosity, tortuosity, and connectivity of vegetation soils. Therefore, it is necessary to investigate the relationship between root length density and gas permeability of vegetated soil.

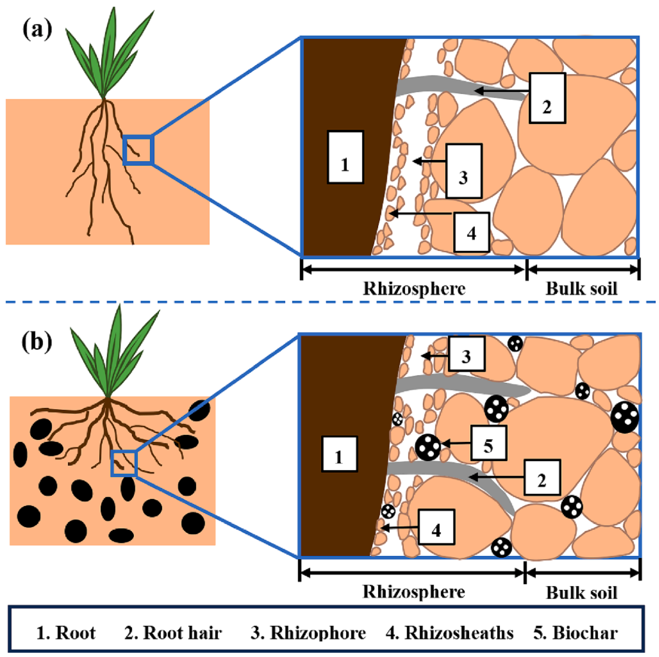


Fig. 8. Unsaturated gas permeability function of biochar-amended vegetated soils based on **(a)** matric suction and **(b)** degree of saturation.

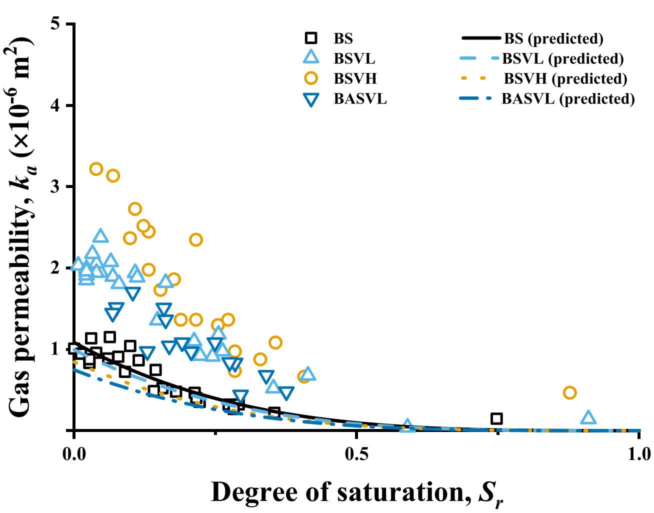


Fig. 9. Prediction of gas permeability function based on the model proposed by Feng et al. (2023).

Table 7
The parameters of gas permeability prediction model proposed by Feng et al. (2023).

Specimen ID	η	R_V	α_r	X_r	RMSE
BS	0.4102	0	2038.2937	3.4976	12.67
BSVL	0.4059	0.0128	2038.2940	3.5601	77.19
BSVH	0.3997	0.0308	2038.2940	3.6903	160.12
BASVL	0.4056	0.0272	2038.2940	3.6643	97.08

Fig. 11 is a permeability curve for a water–gas system in soils. The k_w with respect to ψ were predicted using a hydraulic conductivity model derived by van Genuchten (1980). The model parameter values were derived from the fitting parameters of SWCC model shown in Table 5. The k_w of rooted soil increased with the increment of root density, which were always larger than the k_w of bare soil in the whole range of suction.

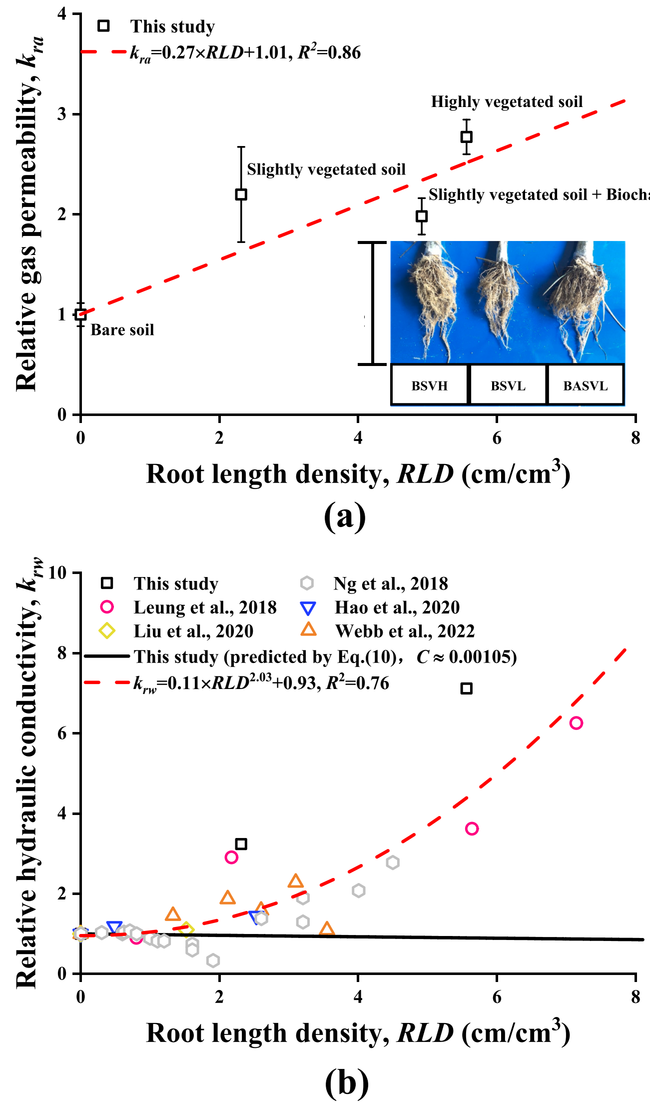


Fig. 10. Dependency of gas permeability on root volume.

Compared with k_s of bare soil ($2.63*10^{-4}$ cm/s), the k_s of BSVL and BSVH were raised by 227 % and 619 % respectively. The enhancement of permeability in vegetated soil, both wetting phases (e.g., water) and non-wetting phases (e.g., gas), would be attributed to the root-induced preferential flow channel (Johnson et al., 2006). Comparing BSVL and BASVL, the addition of biochar reduced k_s by 33 %. This may be because biochar blocked the preferential flow pathway. The k_w stayed rather high for a long time at low suction and dropped later when the suction reached air entry value. Unlike hydraulic conductivity which maintained a maximum value at low suction, the gas cannot percolate through the soil until the suction exceeded the air entry value. This was because the capillary pressure made water films block up enough of the soil pore system (Richards et al., 1931). Even though there may be some gas in the soil with the increase of suction, the gas was isolated to not participate in transport in the suction ranges from 1 kPa to air entry value. The curve shapes shown in Fig. 11 demonstrated that the total permeability under funicular regime, $k_w + k_a$, was less than that in the region of only one phase flow (i.e., capillary regime and pendular regime). The interfacial coupling effects of two phases would strengthen the resistance of flow in soil, which was larger than that of single-phase flow in soil (Zhao et al., 2018).

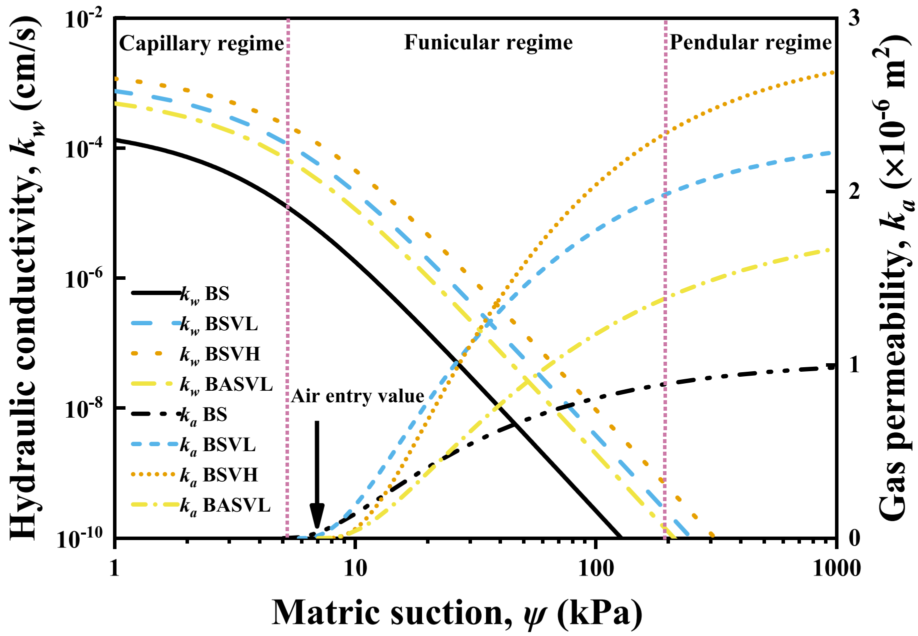


Fig. 11. Water and air permeability functions of biochar-amended vegetated soils.

4. Conclusions

The current study explored the feasibility of using biochar to decrease the unwanted increase of CO_2 permeability in vegetated landfill covers. The variation in CO_2 gas permeability was established with respect to soil–water characteristics and root length density. The following major conclusions can be drawn from the current study:

1. Roots enhanced the fluid flow (including gas and water) in soil by increasing the gas permeability and hydraulic conductivity by up to 3 times and 7 times, respectively. This may be attributed to the preferential flow pathway formed at the root-soil interface (i.e., rhizophore). The suction of BSVH reached up by 50 % compared with BS. This might be due to the increasement of micropore caused by the extrusion of thick roots and occupation of fine roots, and the negative pressure caused by root water absorption.
2. Biochar facilitated plant growth by raising root volume and shoot height by around 110 % and 20 %, respectively. This might be attributed to water retention and nutrient supply of biochar. It was found that the biochar addition would decrease the gas permeability and hydraulic conductivity around 21 % and 33 %, respectively.
3. A model to predict gas permeability correlated with unsaturated soil properties was proposed. The accurate gas permeability of vegetated biochar-amended soil could not be obtained through calibrating the porosity and connectivity of bare soil. In vegetated soils, incorporation of root length density, which approximately idealizes the length of preferential flow channels provided a better prediction of actual CO_2 gas permeability.

CRediT authorship contribution statement

X. Huang: Writing – original draft, Formal analysis, Data curation, Conceptualization. **W. Cai:** Writing – review & editing, Investigation, Data curation, Conceptualization. **S. Bordoloi:** Writing – review & editing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

The last author acknowledge the Tenure Track fund provided by Aalto University.

References

Abiven, S., Hund, A., Martinsen, V., Cornelissen, G., 2015. Biochar amendment increases maize root surface areas and branching: a shovelomics study in Zambia. Plant Soil. 395, 45–55. https://doi.org/10.1007/s11104-015-2533-2.
Astm d2487–17, 2020. Standard practice for classification of soils for engineering purposes (Unified Soil Classification System), ASTM. West Conshohocken. https://doi.org/10.1520/D2487-17.
Bodner, G., Leitner, D., Kaul, H.P., 2014. Coarse and fine root plants affect pore size distributions differently. Plant Soil. 380, 133–151. https://doi.org/10.1007/s11104-014-2079-8.
Bordoloi, S., Ng, C.W.W., 2020. The effects of vegetation traits and their stability functions in bio-engineered slopes: A perspective review. ENG GEOL. 275, 105742 https://doi.org/10.1016/j.enggeo.2020.105742.
Brooks, R.H. and Corey, A.T., 1964. Hydraulic Properties of Porous Media. Hydrology Papers, No. 3, Colorado State U., Fort Collins, Colorado. http://hdl.handle.net/10217/61288.
Bruun, E.W., Petersen, C.T., Hansen, E., Holm, J.K., Hauggaard-Nielsen, H., 2014. Biochar amendment to coarse sandy subsoil improves root growth and increases water retention. Soil Use Manag. 30 (1), 109–118. https://doi.org/10.1111/sum.12102.
Cai, W., Bordoloi, S., Ng, C.W.W., Sarmah, A.K., 2022. Influence of pore fluid salinity on shrinkage and water retention characteristics of biochar amended kaolin for landfill liner application. Sci. Total Environ. 838, 156493 https://doi.org/10.1016/j.scitotenv.2022.156493.
Chen, R., Huang, J.W., Leung, A.K., Chen, Z.K., Ping, Y., Xu, Y., 2023. Modelling air conductivity function of unsaturated root-permeated soil. Soil Tillage Res. 227, 105583 https://doi.org/10.1016/j.still.2022.105583.
Delahaye, C.H., Alonso, E.E., 2002. Soil heterogeneity and preferential paths for gas migration. Eng Geol. 64 (2–3), 251–271. https://doi.org/10.1016/S0013-7952(01)00104-1.
Feng, S., Huang, S.F., Ng, C.W.W., Chen, F.Q., Qian, X., Zhao, N.K., 2023. Roots of Cynodon dactylon increase gas permeability and gas diffusion coefficient of highly compacted soils. Plant and Soil 492 (1), 329–351.
Gallipoli, D., Wheeler, S.J., Karstunen, M., 2003. Modelling the variation of degree of saturation in a deformable unsaturated soil. Géotechnique 53 (1), 105–112. https://doi.org/10.1680/geot.2003.53.1.105.

Garg, A., Bordoloi, S., Ni, J., Cai, W., Maddibiona, P.G., Mei, G., Poulsen, T.G., Lin, P., 2019. Influence of biochar addition on gas permeability in unsaturated soil. Geotech. Lett. 9 (1), 66–71. https://doi.org/10.1680/jgele.18.00190.

Garg, A., Huang, H., Kushvaha, V., Madhushri, P., Kamchoom, V., Wani, I., Koshy, N., Zhu, H.H., 2020. Mechanism of biochar soil pore–gas–water interaction: gas properties of biochar-amended sandy soil at different degrees of compaction using KNN modeling. Acta Geophys. 68, 207–217. https://doi.org/10.1007/s11600-019-00387-y.

Ghestem, M., Sidle, R.C., Stokes, A., 2011. The influence of plant root systems on subsurface flow: implications for slope stability. Bioscience. 61 (11), 869–879. https://doi.org/10.1525/bio.2011.61.11.6.

Gill, S.E., Handley, J.F., Ennos, A.R., Pauleit, S., 2007. Adapting cities for climate change: the role of the green infrastructure. Build Environ. 33 (1), 115–133. https://doi.org/10.2148/benv.33.1.115.

Guo, H., Zhang, Q., Chen, Y., Lu, H., 2023. Effects of biochar on plant growth and hydro-chemical properties of recycled concrete aggregate. Sci. Total Environ. 882, 163557 https://doi.org/10.1016/j.scitotenv.2023.163557.

Hu, X., Li, X.Y., Wang, P., Liu, Y., Wu, X.C., Li, Z.C., Zhao, Y.D., Cheng, Y.Q., Guo, L.L., Lyu, Y.L., Liu, L.Y., 2019. Influence of exclosure on CT-measured soil macropores and root architecture in a shrub-encroached grassland in northern China. Soil Tillage Res. 187, 21–30. https://doi.org/10.1016/j.still.2018.10.020.

Jeffery, S., Meinders, M.B., Stoof, C.R., Bezemer, T.M., van de Voorde, T.F., Mommer, L., van Groenigen, J.W., 2015. Biochar application does not improve the soil hydrological function of a sandy soil. Geoderma. 251, 47–54. https://doi.org/10.1016/j.geoderma.2015.03.022.

Johnson, M.S., Lehmann, J., 2006. Double-funneling of trees: Stemflow and root-induced preferential flow. Ecoscience. 13 (3), 324–333. https://doi.org/10.2980/i1195-6860-13-3-324.1.

Joseph, S., Cowie, A.L., Van Zwieten, L., Bolan, N., Budai, A., Buss, W., Cayuela, M.L., Graber, E.R., Ippolito, J.A., Kuzyakov, Y., Luo, Y., Ok, Y.S., Palansooriya, K.N., Shepherd, J., Stephens, S., Weng, Z., Lehmann, J., 2021. How biochar works, and when it doesn't: A review of mechanisms controlling soil and plant responses to biochar. GCB Bioenergy. 13 (11), 1731–1764. https://doi.org/10.1111/gcbb.12885.

Juca, J. F. T., Maciel, F. J., 2006. Gas permeability of a compacted soil used in a landfill cover layer, in: Miller, G.A., Zapata, C.E., Houston, S.L., Fredlund, D.G. (Eds.), Unsaturated soils 2006, United States, pp. 1535-1546. https://doi.org/10.1061/40802(189)128.

Kaza, S., Yao, L., Bhada-Tata, P.,Van Woerden, F., 2018. What a waste 2.0: a global snapshot of solid waste management to 2050. Washington, DC: World Bank. http://hdl.handle.net/10986/30317.

Keiluweit, M., Nico, P.S., Johnson, M.G., Kleber, M., 2010. Dynamic molecular structure of plant biomass-derived black carbon (biochar). Environ. Sci. Technol. 44 (4), 1247–1253. https://doi.org/10.1021/es9031419.

Koebernick, N., Daly, K.R., Keyes, S.D., George, T.S., Brown, L.K., Raffan, A., Cooper, L. J., Naveed, M., Bengough, A.G., Sinclair, I., Hallett, P.D., Roose, T., 2017. High-resolution synchrotron imaging shows that root hairs influence rhizosphere soil structure formation. New Phytol. 216 (1), 124–135. https://doi.org/10.1111/nph.14705.

Kolb, E., Legué, V., Bogeat-Triboulot, M.B., 2017. Physical root–soil interactions. Phys Biol. 14 (6), 065004 https://doi.org/10.1088/1478-3975/aa90dd.

Lee, Y., Ryu, C., Park, Y.K., Jung, J.H., Hyun, S., 2013. Characteristics of biochar produced from slow pyrolysis of Geodae-Uksae 1. Bioresour. Technol. 130, 345–350. https://doi.org/10.1016/j.biortech.2012.12.012.

Lehmann, J., Rillig, M.C., Thies, J., Masiello, C.A., Hockaday, W.C., Crowley, D., 2011. Biochar effects on soil biota–a review. Soil Biol. Biochem. 43 (9), 1812–1836. https://doi.org/10.1016/j.soilbio.2011.04.022.

Lu, J., Zhang, Q., Werner, A.D., Li, Y., Jiang, S., Tan, Z., 2020. Root-induced changes of soil hydraulic properties–A review. J. Hydrol. 589, 125203 https://doi.org/10.1016/j.jhydrol.2020.125203.

Mitchell, J.K., Soga, K., 2005. Fundamentals of Soil Behavior Vol. 3, p..

Ni, J.J., Leung, A.K., Ng, C.W.W., 2019. Modelling effects of root growth and decay on soil water retention and permeability. Can. Geotech. J. 56 (7), 1049–1055. https://doi.org/10.1139/cgj-2018-0402.

Peng, Y., Niu, J., Peng, Z., Zhang, F., Li, C., 2010. Shoot growth potential drives N uptake in maize plants and correlates with root growth in the soil. Field Crops Res. 115 (1), 85–93. https://doi.org/10.1016/j.fcr.2009.10.006.

Richards, L.A., 1931. Capillary conduction of liquids through porous mediums. Physics. 1 (5), 318–333. https://doi.org/10.1063/1.1745010.

Sarmah, A.K., 2021. Biochar: fundamentals and applications in environmental science and remediation technologies. Academic Press.

Shafawi, A.N., Mohamed, A.R., Lahijani, P., Mohammadi, M., 2021. Recent advances in developing engineered biochar for CO2 capture: An insight into the biochar modification approaches. Journal of Environmental Chemical Engineering 9 (6), 106869. https://doi.org/10.1016/j.jece.2021.106869.

Shaikh, J., Bordoloi, S., Yamsani, S.K., Sekharan, S., Rakesh, R.R., Sarmah, A.K., 2019. Long-term hydraulic performance of landfill cover system in extreme humid region: Field monitoring and numerical approach. Sci. Total Environ. 688, 409–423. https://doi.org/10.1016/j.scitotenv.2019.06.213.

Shao, W., Ni, J., Leung, A.K., Su, Y., Ng, C.W.W., 2017. Analysis of plant root–induced preferential flow and pore-water pressure variation by a dual-permeability model. Can. Geotech. J. 54 (11), 1537–1552. https://doi.org/10.1139/cgj-2016-0629.

Tarafdar, J.C., Jungk, A., 1987. Phosphatase activity in the rhizosphere and its relation to the depletion of soil organic phosphorus. Soil Biol. Biochem. 3, 199–204. https://doi.org/10.1007/BF00640630.

Tavanti, R.F.R., Liñares, M.L., Soares, M.B., Trevisan, R.G., Tavanti, T.R., da Silva Freddi, O., Montanari, R., González, A.P., 2023. Macro-scale spatial modeling reveals the role of soil organic matter quality in CO2 emissions. Geoderma Regional 34, e00690.

Van Genuchten, M.T., 1980. A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. Soil Sci Soc Am J. 44 (5), 892–898. https://doi.org/10.2136/sssaj1980.03615995004400050002x.

Wang, F., Chen, H., Lian, J., Fu, Z., Nie, Y., 2018. Preferential flow in different soil architectures of a small karst catchment. Vadose Zone J. 17 (1), 1–10. https://doi.org/10.2136/vzj2018.05.0107.

Wang, M.Y., Siddiqi, M.Y., Glass, A.D.M., 1996. Interactions between K+ and NH4+: effects on ion uptake by rice roots. Plant Cell Environ. 19 (9), 1037–1046. https://doi.org/10.1111/j.1365-3040.1996.tb00210.x.

Wong, J.T.F., Chen, Z., Ng, C.W.W., Wong, M.H., 2016. Gas permeability of biochar-amended clay: potential alternative landfill final cover material. Environ. Sci. Pollut. Res. 23, 7126–7131. https://doi.org/10.1007/s11356-015-4871-2.

Xiao, T., Li, P., Fei, W., Wang, J., 2023. Effects of vegetation roots on the structure and hydraulic properties of soils: A perspective review. Sci. Total Environ. 167524 https://doi.org/10.1016/j.scitotenv.2023.167524.

Zhao, J., Kang, Q., Yao, J., Viswanathan, H., Pawar, R., Zhang, L., Sun, H., 2018. The effect of wettability heterogeneity on relative permeability of two-phase flow in porous media: A lattice Boltzmann study. Water Resour. Res. 54 (2), 1295–1311. https://doi.org/10.1002/2017WR021443.

Zhao, B., Santamarina, J.C., 2020. Desiccation crack formation beneath the surface. Géotechnique. 70 (2), 181–186. https://doi.org/10.1680/jgeot.18.T.019.